# Breast Cancer Prediction: Impact of Stratified Sampling Approach on Classifier Accuracy

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***Abstract-*** **Late identification of breast cancer is a major concern in India. After lung cancer, breast cancer is the second most common cause of mortality for women. In the present study, artificial intelligence is utilised to determine if a breast tumour is benign or malignant. The goal of this paper is to offer a thorough investigation of how K Nearest Neighbor (K-NN) is used to identify breast cancer. One of the most basic supervised learning-based machine learning algorithms is K-NN. Additionally, we have investigated how stratified sampling affects the classifier's accuracy. The proposed work shows that stratified sampling increases accuracy and F1-scores considerably.**

***Keywords: K Nearest Neighbor Classifier, breast cancer, stratified sampling, accuracy***

## I. Introduction

Breast tissue can become cancerous and evolve into breast cancer. Breast cancer is a malignant tumour that develops in breast cells. The possibility of a tumour spreading to different body regions exists. A lump in the breast, a change in the size and form of the breast, dimpling of the skin, a newly inverted nipple, or a red or scaly patch of skin can all be indicators of breast cancer. There may be bone pain, swollen lymph nodes, shortness of breath, or yellow skin when the disease has gone far and wide.

Machine learning has taken on significant importance in modern society because it offers clever and reasonably priced answers to a range of issues. As a result, healthcare is attracting the attention of many researchers because society depends on healthy, productive people for its well-balanced operation. Machine learning (ML) techniques have been applied by several researchers to solve health care problems. Machine learning tasks are usually described in terms of how the machine learning system should process an example, or experience, or data point. Machine learning tasks mainly include Classification, Regression, clustering, anomaly detection to name a few. To evaluate a model, a performance measure is required, and accuracy of the model is a very commonly used one. Accuracy is the proportion of data instances for which the model produces the correct output. A broad categorization of machine learning is into Supervised and Unsupervised, depending on whether the model is given labelled or unlabelled dataset to train from. By using the collected dataset, the models are trained and when the test data arrives, the prediction is obtained by applying the model. Likewise, ML approaches have been applied for diagnosis of breast cancer. The early diagnosis of breast cancer has taken a new turn thanks to machine learning algorithms. One of the most challenging issues in medical science is the identification of breast cancer, and machine learning can be very helpful in this area. This issue has been the focus of numerous research projects. Authors such as Seyyid et al. in [1], and Priya et al. in [2] have applied the K Nearest Neighbour algorithm for breast cancer prediction. Victoria et al. also applied this algorithm with different distance measurements [3]. The experiments by these researchers proved that good results can be obtained with the application of KNN algorithm. Assegie applied the KNN algorithm and optimized it by parameter tuning to get an accuracy of 94.35% as compared to 90.10% obtained with default parameters [4]. Authors in [5] [6] [8] [10] have applied KNN algorithm for Breast cancer classification. An unsupervised approach has been applied by Aidarus et al [7] for diagnosing breast cancer. Various machine learning techniques have been used by other researchers [9] to create disease prediction models. Moreover, due to the large amount of healthcare data which is being collected, study of machine learning techniques in the field of Big data for healthcare has also been an active area of research [12][13][14][15][16]. The idea behind this study is that the key to treating breast cancer is early detection. The likelihood of survival increases with breast cancer stage. Moreover, proper data handling is required in order to create stronger classification models, as demonstrated in this article via the use of stratified sampling.

## II. Methods

K Nearest Neighbour- This is a very simple algorithm, and is often called a lazy learner algorithm. It applies the fact that there are similarity in data points if they belong to the same class. The algorithm stores all of the available data, and then uses similarity to categorise fresh data points. When a test data arrives, it can be quickly categorised into a suitable category using the K-NN algorithm. Although the K-NN technique is utilised for both classification and regression problems, it is mostly employed for classification tasks. The graphical representation of KNN algorithm is shown in fig. 1.The working of KNN is briefly described in the following steps: selection of K; computing the distances of test data with all datapoints and selecting the K nearest ones; among these K neighbours, the number of the data points in each category is counted; finally the test datapoint is assigned to that category for which the number of the neighbours is maximum.



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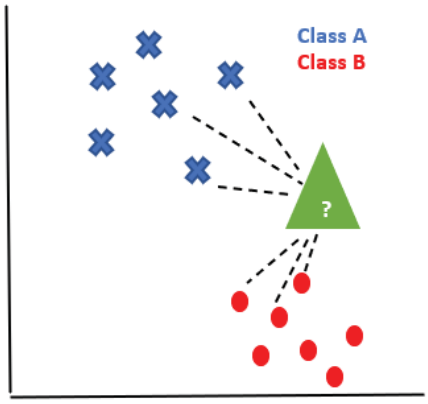


Fig. 1. Graphical representation of KNN algorithm

### A. Stratified Sampling

The population is divided into smaller subgroups known as strata as part of the sampling procedure known as stratified random sampling. The strata are established in stratified random sampling or stratification based on the shared characteristics or attributes of the members, such as health or educational attainment. Random samples are drawn from stratified groups proportional to the population in proportional stratified random sampling. On the other hand, in disproportionate sampling, strata are not proportional to the distribution of the population.

Stratified random sampling assures that each prospective sample has an equal chance of occuring, in contrast to basic random sampling, which chooses data at random from a population as a whole. Proportionate stratified sampling has been used in this study project.

## III. Experiments

### A. Simulation Tools

The simulation tools used here are mentioned as follows:

1) *Python 3.9*
2) *IDE – Jupyter Notebook*
3) *Python packages*
   a) *Pandas*
   b) *NumPy*
   c) *Matplotlib*
   d) *Scikit-learn*

### B. Dataset:

This study has used the University of Wisconsin's WBCD (Wisconsin breast cancer database) [11]. The database includes breast cancer data from human breast tissue. Experiments were conducted with varying values of the parameter K. The dataset description is given in Table 1 and class-wise graphical distribution is shown in Fig. 2.

Dataset Description

| Class | WBCD |
|---|---|
| 0 (benign tumour) | 357 |
| 1 (malignant tumour) | 212 |

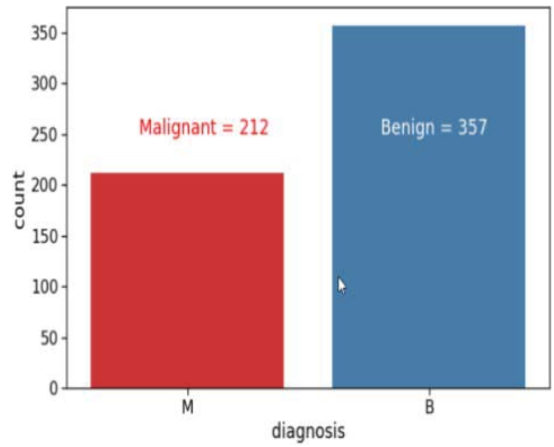


Fig. 2. Graphical representation of class wise distribution

### C. Evaluation Parameter

Accuracy and f1 score were employed as the evaluation criteria for this research project. Accuracy of a dataset is equal to the total number of correctly classified examples divided by the dataset's total number of examples.

The harmonic mean of recall and precision is the f1 score. By combining accuracy and recall, two metrics that would normally be in competition, it elegantly summarises the prediction ability of a model.

## IV. Results

The program was run multiple times using different values of K, different ratio of training and test data. We have also conducted experiments applying stratified sampling. The results are shown in Fig. 3 – 5.

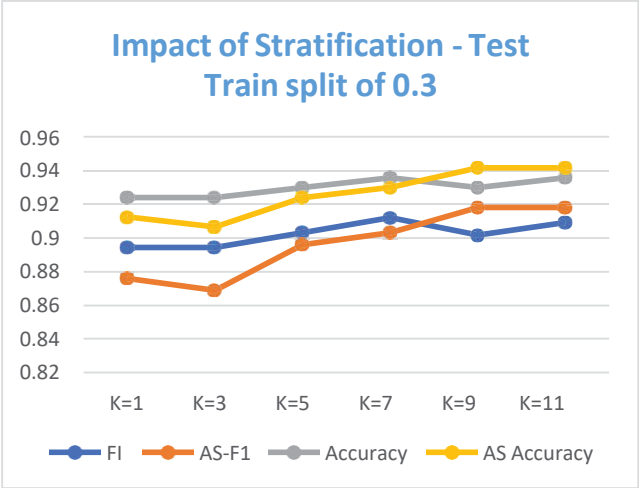


Fig. 3. Accuracy and F1-Scores before and after stratification (AS) for Test Train split of 0.2



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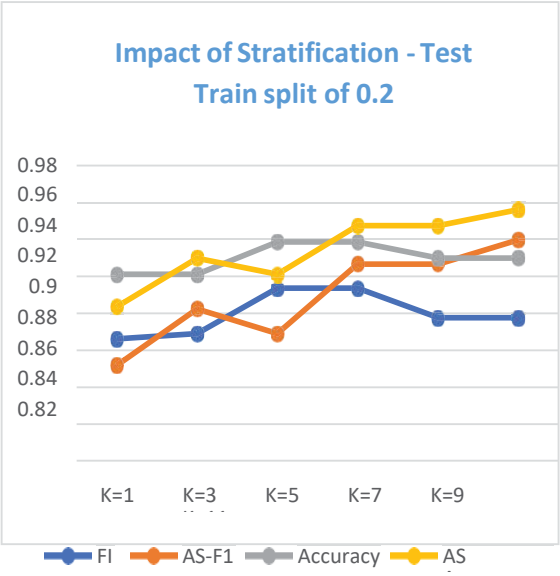


Fig. 4. Accuracy and F1-Scores before and after stratification (AS) for Test Train split of 0.3

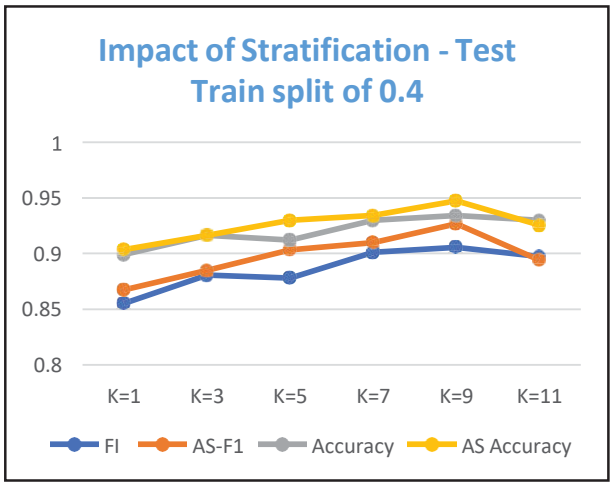


Fig. 5. Accuracy and F1-Scores before and after stratification (AS) for Test Train split of 0.4

## V. Analysis

The experiments were conducted with the different values of K, varying Test Train splits, and with and without stratified sampling.

For the Test Train split of 0.2, it is seen from Figure 2, that except for K = 1 and K = 5, the Accuracy and F1-Score values increase after stratification. For the Test Train split of 0.3, it is seen from Figure 3, that except for K = 9 and K

= 11, the Accuracy and F1-Score values decrease after stratification. For the Test Train split of 0.4, it is seen from Fig. 4, that for all values of K, the Accuracy and F1- Score values improve after stratification.

## VI. Conclusion

Machine learning helps in the building of disease prediction models and has been applied in early diagnosis of breast cancer by several researchers. The diagnosis of breast cancer is one of the most difficult problems in medical science and machine learning offers great help in this direction. This work gives a comprehensive analysis of the usage of K Nearest Neighbour (K-NN) to diagnose breast cancer. We have conducted experiments with proportionate stratified sampling for different K values and Test Train splits. The highest accuracy score of 95.61% and F1-Score of 93.98% have been obtained. Several researchers have worked to find solutions to this problem. However, our work is novel because we have analyzed the impact of stratification of data used for building the model. We further aim to apply more efficient models with representation learning to build efficient models for disease prediction.

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